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0000 separate (IDL.SEM_PHASE)
NUM (TXT : String) return TREE is
use UNIV_OPS;
MAXCOL : constant Integer := TXT'LAST; COL : Integer := TXT'FIRST; CHR : Character :=
:= TXT (TXT'FIRST); SCALEFACTOR : Integer := Integer'LAST; VNUMBER : VECTOR; VDENOM : VECTOR; RADIX : Integer :=
10; EXPONENT : Integer := 0; EXPONENT_POSITIVE : Boolean := True; procedure V_CLEAR (ARG : out VECTOR) is begin ARG :=
:= +1; ARG.L := 1; ARG.D(1) := 0; end V_CLEAR; procedure V_INCREMENT (DLTA : Integer; V : in out VECTOR) is begin ARG :=
CARRY : UDIGIT := 0; begin V_SCALE (RADIX, V); TEMP := V.D(1) + UDIGIT (DLTA); if TEMP >= 10_000 then CARRY := 1; TEMP :=
= TEMP - 10_000; end if; V.D(1) := TEMP; for I in 2 .. V.L loop if CARRY = 0 then return; end if; TEMP := V.D(I) +
I) + 1; if TEMP >= 10_000 then CARRY := 1; TEMP := TEMP - 10_000; else CARRY := 0; end if; V.D(I) :=
TEMP; end loop; if CARRY > 0 then V.L := V.L + 1; V.D(V.L) := CARRY; NORMALIZE (V); -- TO CHECK FOR OVERFLOW end
if; end V_INCREMENT; procedure NEXT_CHR is begin COL := COL + 1; if COL <= MAXCOL then CHR := TXT (COL); else CHR := '
'; end if; end NEXT_CHR; begin V_CLEAR (VNUMBER); -- GET INTEGER VAL OR RADIX while CHR in '0' .. '9' or else CHR = '.' or else CHR
= ' ' loop if CHR = '.' then SCALEFACTOR := 0; elsif CHR = '-' then V_INCREMENT (Character'POS (CHR) - Character'POS ('0'), VNUME
R); SCALEFACTOR := SCALEFACTOR - 1; end if; NEXT_CHR; end loop; if CHR = '#' then RADIX := Integer (VNUMBER.D (1)); VN
UMER.D (1) := 0; NEXT_CHR; while CHR /= '#' loop if CHR = '.' then SCALEFACTOR := 0; elsif CHR = ' ' then null;
elsif CHR <= '9' then V_INCREMENT (Character'POS (CHR) - Character'POS ('0'), VNUMBER); SCALEFACTOR := SCALEFACTOR - 1; else
V_INCREMENT (Character'POS (CHR) - Character'POS ('A') + 10, VNUMBER); SCALEFACTOR := SCALEFACTOR - 1; end if; NEXT_CHR;
end loop; NEXT_CHR; end if; if CHR = 'E' then NEXT_CHR; if CHR = '+' then NEXT_CHR; elsif CHR = '-' then NEXT_CHR;
EXPONENT_POSITIVE := False; end if; while CHR /= ' ' loop if EXPONENT >= 3_275 then Put_Line ("!! EXPONENT IN NUMERIC LIT T
OO LARGE"); raise Program_Error; end if; EXPONENT := EXPONENT * 10 + Character'POS (CHR) - Character'POS ('0'); NEXT_CHR;
end loop; if not EXPONENT_POSITIVE then EXPONENT := -EXPONENT; end if; end if; if SCALEFACTOR < 0 then EXPONENT := EXPONENT +
0020 SCALEFACTOR; V_CLEAR (VDENOM); VDENOM.D (1) := 1; end if; if EXPONENT > 0 then for I in 1 .. EXPONENT loop V_SCALE (RADIX, VNUM
ER); end loop; elsif EXPONENT < 0 then for I in 1 .. -EXPONENT loop V_SCALE (RADIX, VDENOM); end loop; end if; if SCALEFACTOR
>= 0 then VNUMBER.D (VNUMBER.L + 1) := 0; return U_INT (VNUMBER); else V_LOWEST_TERMS (VNUMBER, VDENOM); VNUMBER.D (VNUMBER.L + 1) := 0;
VDENOM.D (VDENOM.L + 1) := 0; return U_REAL (VNUMBER, VDENOM); end if; end EVAL_NUM;

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